

Advanced irrational equations

№1. Solve an equation:

$$\sqrt{\sqrt{x+2} + \sqrt{x-2}} = 2\sqrt{\sqrt{x+2} - \sqrt{x-2}} + \sqrt{2}.$$

9290'91

№2. Solve an equation:

$$\frac{1}{\sqrt{x+2} + \sqrt{x+3}} + \frac{1}{\sqrt{x+3} + \sqrt{x+4}} + \dots + \frac{1}{\sqrt{x+2017} + \sqrt{x+2018}} = 42.$$

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№3. Solve an equation:

$$\sqrt{x - 2\sqrt{x-2} - 1} + \sqrt{x - 2\sqrt{x-3} - 2} = \sqrt{x-2}.$$

7'3

№4. Solve an equation:

$$\sqrt[3]{15x+1-x^2} + \sqrt[3]{x^2-15x+27} = 4.$$

0;2;13;13;7;15

№5. Solve an equation:

$$\sqrt{x-15} + \sqrt{x+80} + 2\sqrt{x-15}\sqrt{x+80} = 315 - 2x.$$

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№6. Find all values of a for which solutions of the equation below exist and provide those solutions:

$$\sqrt{x - \sqrt{3}} + a^2x^2 + 2ax(\sqrt{6} - \sqrt{3}) = 6\sqrt{2} - 9.$$

8^ = x '2^ - 1 = v